



Detailed Explanation of Computer Hardware Components



1. SMPS (Switched Mode Power Supply)

Function:

Converts AC (Alternating Current) from the wall socket into DC (Direct Current) needed by computer components.

Key Features:

Supplies power to motherboard, hard drives, GPU, etc.

Comes in different wattage (e.g., 450W, 600W, 750W).

Efficiency ratings: 80 Plus, Bronze, Gold, Platinum (higher is better).

Connectors:

24-pin ATX connector – to motherboard

4/8-pin CPU connector – to processor

PCIe connectors – for graphics card

SATA/Molex – for storage devices



SMPS (ATX Power Supply) Wire Color Codes

Wire Color	Voltage / Signal	Purpose
Yellow	+12 V	CPU, GPU, motors, HDD/SSD
Red	+5 V	USB, logic circuits
Orange	+3.3 V	RAM, chipset
Black	Ground (0 V)	Return / Earth
Green	PS_ON	Power ON signal (short to GND to start SMPS)
Purple	+5 V Standby (5VSB)	Always ON power

Gray	PWR_OK	Power good signal
Blue	-12 V	Legacy serial ports
White	-5 V	Old AT power supplies (obsolete)
Brown	+3.3 V Sense	Voltage sensing
Pink	+5 V Sense	Voltage sensing (rare)

+3.3V	1	11	+3.3V
+3.3V	2	12	-12V
Common	3	13	Common
+5V	4	14	PS On
Common	5	15	Common
+5V	6	16	Common
Common	7	17	Common
Pwr OK	8	18	-5V
+5V SB	9	19	+5V
+12V	10	20	+5V



Basic Wire Color Coding & Testing +3.3V, +5V, +12V and -12V

Key Checks (Troubleshooting)

- ◆ **System Not Powering ON**
 - ✓ Check Purple wire (5VSB)
 - ✗ No 5V → SMPS faulty
- ◆ **SMPS Not Starting**
 - ✓ Short Green + Black
 - ✗ Fan not spinning → Dead SMPS
- ◆ **Random Shutdown / Restart**
 - ✓ Check Gray (PWR_OK)
 - ✗ Low or fluctuating → Faulty SMPS



2. RAM (Random Access Memory)

Function:

Temporary storage used by the CPU to run programs and process data.

Characteristics:

Volatile memory (data is lost when power is off).

Speed is measured in MHz (e.g., DDR4 3200MHz).

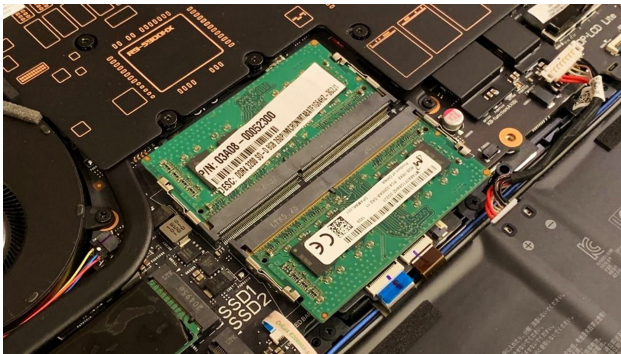
Capacity: Common sizes – 4GB, 8GB, 16GB, 32GB.

Types:

DDR3, DDR4, DDR5 (latest, faster, and more efficient).

Usage:

More RAM = smoother multitasking and faster applications.



DDR



DDR2



DDR3



DDR4



⚙️ 3. Processor (CPU – Central Processing Unit)

Function:

The "brain" of the computer, it performs calculations and executes instructions.

Components:

ALU (Arithmetic Logic Unit): Handles arithmetic and logic.

CU (Control Unit): Directs operation of the processor.

Cache: High-speed memory inside CPU for quick data access.

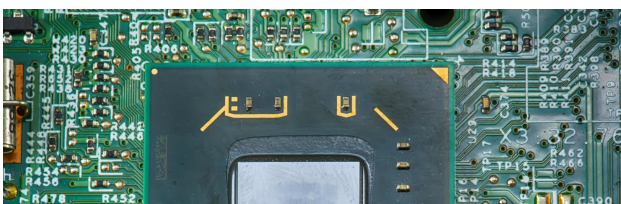
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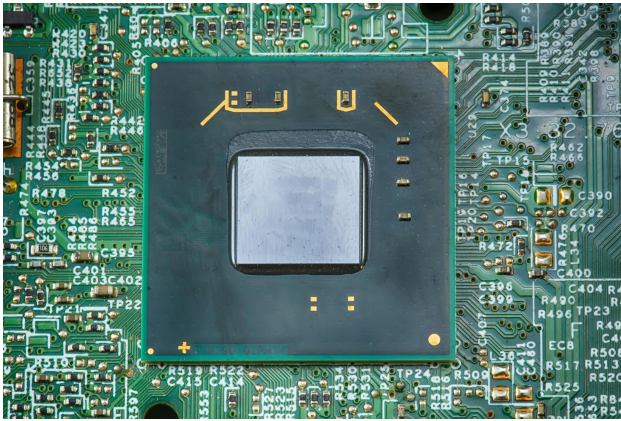
Cores & Threads: More cores = better multitasking.

E.g., Intel i5 (6 cores), AMD Ryzen 7 (8 cores)

Clock Speed: Measured in GHz (e.g., 3.6GHz)

Architecture: 64-bit, x86/x64





- i3 : Entry Level Processor (Basic): Documentation/Browsing (2-4 Cores/)
- i5: Mid-range
- i7- High performance
- i9- Extreme performance

✚ 4. Ports

Function:

Used to connect external devices like keyboard, mouse, monitor, etc.

Common Ports:

Port Type	Description
USB (Type-A/C)	Connects peripherals (mouse, pen drive)
HDMI	Transfers HD video/audio to monitor/TV
VGA	Analog video output
Ethernet (RJ-45)	Wired network connectivity
Audio Jack	Sound input/output
DisplayPort	High-res display interface (like HDMI)
PS/2	Legacy keyboard/mouse connection



🎮 5. Graphics Card (GPU – Graphics Processing Unit)

Function:

Processes and renders images, videos, animations, and 3D graphics.

Types:

Integrated GPU: Built into CPU (e.g., Intel UHD, AMD Vega).

Dedicated GPU: Separate component (e.g., NVIDIA GeForce, AMD Radeon).

Key Features:

VRAM (e.g., 4GB, 8GB GDDR6)

Useful for gaming, video editing, AutoCAD, machine learning.

Slots Used:

PCIe x16 slot on motherboard

Requires separate power from SMPS (6-pin, 8-pin)



6. Motherboard

Function:

Main circuit board that connects and allows communication between all components.

Main Parts:

CPU Socket: Where processor is installed

RAM Slots (DIMM): For installing memory

Chipset: Controls communication between CPU, RAM, and peripherals

PCIe Slots: For expansion cards (graphics, Wi-Fi)

SATA/M.2 Ports: Connect storage devices (HDD, SSD)

I/O Panel: USB, audio, Ethernet ports

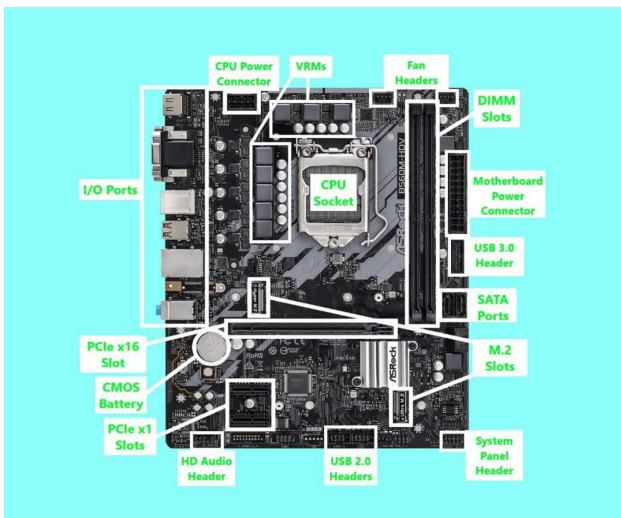
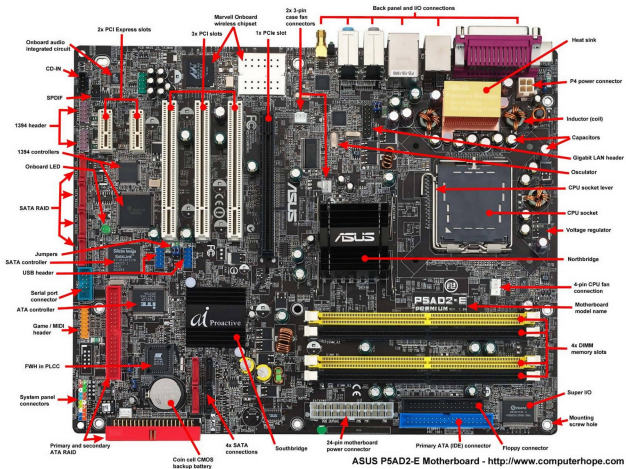
BIOS/UEFI Chip: Firmware that helps boot the system

Form Factors:

ATX (Full size)

Micro-ATX

Mini-ITX



◆ What is HDD (Hard Disk Drive)?

- HDD is an **electromechanical storage device**.
- It uses **spinning magnetic disks (platters)** to read/write data.
- **Slower** than SSD but **cheaper** for high-capacity storage.

Features:

- **Mechanical moving parts** (spindle, actuator arm).
- **Large storage** (500GB to several TB).
- Common in **desktops and older laptops**.

◆ What is SSD (Solid State Drive)?

- SSD is a **faster** storage device using **flash memory** (no moving parts).
- Boots OS and loads software **much faster** than HDD.

Features:

- **No mechanical parts** – uses NAND flash.
- **Faster read/write speeds.**
- More **durable**, **silent**, and **energy-efficient**.
- Common in **modern laptops and PCs**.

